

Long Branch Remover

Formulas and decision rules as implemented

A reference for what the code actually computes. Every formula and every condition below is what is in `long_branch_rules.jl`, in the order the code applies it. The reasoning behind the choices is in `long_branch_python_vs_julia.pdf`; how to run it is in `HOWTO_run.md`.

***Preliminary.** This document was produced with Claude AI during the debugging and revision of the Julia implementation, which was itself carried out with Claude AI. It describes code that is still being tested, and should be treated as a working record rather than a settled specification. Anything here may change as testing continues.*

0. Notation

Symbol	Meaning
K	trigger value for a rule, from the -t file
q	safeguard quantile for a rule, from the -s file
m _u	median of the observations belonging to unit u (a taxon, a clade, or a tree)
M	global constant for that rule: the median of all the per-unit medians
Q(q)	the q-th quantile of the pooled list of every observation for that rule
bl	the branch length being tested

Constant	Value	Meaning
MIN_CROSS_GENE_OBS	3	trees in which a unit must be observed before it can be tested
MIN_COMPARATOR_DISTANCE	1e-6	smallest distance from the anchor at which a comparator carries usable information

Two distributions, used for different things. The per-unit medians give the trigger; the pooled list of every observation gives the safeguard. A unit votes once in M regardless of how often it was sampled, whereas Q(q) is a percentile of real branches and so counts every observation.

1. The common trigger

Three of the four rules share one form:

```
trigger_u = (K - 1) * m_u + M
```

When a unit is exactly typical (m_u equals M) this reduces to K * M, so K reads as "how many times the typical value". Both the trigger and the safeguard must be exceeded before anything is nominated:

```
remove if bl > trigger_u AND bl > Q(q)
```

Both comparisons are strict, so a value sitting exactly on a threshold is retained.

Rule	unit u	m _u is the median of	M is the median of
Terminal	taxon	that taxon's terminal branch across trees	the per-taxon medians
Clade stem	named clade	that clade's stem across trees	the per-clade medians
Internal	gene tree	all internal branches in that tree	the per-tree medians

2. Terminal rule

`identify_long_terminals`. For each leaf in each tree:

```
trigger_taxon = (K1 - 1) * median(bl of this taxon across trees) + M_terminal
```

```
remove if    bl > trigger_taxon    AND    bl > Q_terminal(q1)
```

A taxon observed in fewer than MIN_CROSS_GENE_OBS (= 3) trees is issued no trigger and is therefore never removed by this rule. Its branch lengths still enter the pooled list and the per-unit medians.

3. Named clade stem rule

identify_clades_with_excessively_long_stems. Applied per tree, in four steps.

3.1 Resolve each clade to a branch

resolve_clade_fragments. Take the clade's taxa present in this tree. If they are monophyletic, that is the fragment. If not, the fragment is the largest monophyletic subset, requiring at least 2 taxa. The measured branch is the one subtending that fragment.

3.2 Collapse aliases

Clade names whose fragments are identical in this tree describe one physical branch and are grouped. Within a group:

Purpose	Which name is used
Which threshold tests the branch	the name with the smallest full definition
Which clade is credited in the report	the name with the largest full definition
Which clades record the measurement	all of them
What enters the pooled background	the branch, once

The first two point deliberately in opposite directions: the tightest name gives the comparable median, the widest name is what the branch stands for.

3.3 Test

```
trigger_clade = (K3 - 1) * median(stem of this clade across trees) + M_stem
remove if    stem_bl > trigger_of_tightest_name    AND    stem_bl > Q_stem(q3)
```

A clade whose stem was measurable in fewer than 3 trees gets no trigger and is not testable.

3.4 Suppress nested clades

Removed branches are ordered largest first. A removed branch whose taxa are a subset of one already accepted is dropped from the report: those taxa are already going.

4. Internal branch and long subtree rule

identify_long_branched_substrees. Four gates, in this order. All must pass.

Gate 1 — build the chain

```
trigger_tree = (K2 - 1) * median(internal bl in this tree) + M_internal
an internal branch is LONG if    bl > trigger_tree
a STACK is a connected set of long internal branches sharing internal nodes
keep the stack if    length(stack) >= p
```

Built on the trigger alone: the safeguard is deliberately not applied here, so a modest branch in the middle cannot split one chain into two and push both below -p.

Gate 2 — safeguard at the branch that acts

get_outermost_edge selects from the stack the edge with the most unbalanced split, i.e. the one subtending the smallest clade. That branch is the stem of the clade that would be removed.

```
continue only if    bl_outermost > Q_internal(q2)
```

Gate 3 — side asymmetry

clade_for_removal_by_asymmetry. For each side of that branch, the median distance from the branch out to the tips on that side, with the tested branch excluded. This measures how long each side is internally.

```
side_ratio = max(median_u, median_v) / min(median_u, median_v)
continue only if    side_ratio > r
the side to remove is the one with the LARGER median depth
```

Gate 4 — the removed side must be smaller

```
continue only if    length(candidate side) < length(other side)
```

If the deeper side is the larger one, nothing is pruned at that branch. The rule does not fall back to removing the other side.

What this excludes. A small clade of short-branched taxa sitting beyond a long stem fails gate 3: both sides look similar, or the clade side is the shallower one. There the long branch is the stem, not the clade.

5. Quartet rule

identify_local_long_branches. Entirely within one tree; no cross-gene background is used for the trigger.

5.1 Masking, per tree

detect_mask_groups_for_tree. For each clade in the -x file, keep the largest monophyletic fragment in this tree, minimum 2 taxa. A clade too scattered, or with fewer than 2 taxa present, is not masked in that tree at all.

5.2 Which comparators are allowed

Focal taxon	Comparator	Allowed
not masked	not masked	yes
not masked	in mask group G	no
in mask group G	not masked	yes
in mask group G	same group G	yes
in mask group G	different group G'	no

5.3 Choosing the quartet

Anchors are internal nodes, searched outward from the focal tip level by level, nearest first. At each anchor the branch leading back to the focal tip is dropped and the remaining branches each contribute a group of allowed comparator tips,

sorted by distance from the anchor. A tip closer to the anchor than MIN_COMPARATOR_DISTANCE is not a valid comparator; it gives the local reference no scale.

```
require at least 2 groups and at least 3 available tips
take the closest tip from each group
if that gives more than 3, keep the 3 closest
if fewer than 3, top up from the pooled remainder, closest first
if 3 cannot be assembled, move to the next anchor
```

The first anchor that yields three comparators is used; the search stops there.

5.4 The test

```
d_focal      = distance(anchor, focal)
local_median = median( distance(anchor, comparator) for the 3 comparators )
ratio        = d_focal / local_median
remove if    ratio > K4    AND    bl_focal > Q_terminal(q1)

not evaluable if    local_median <= MIN_COMPARATOR_DISTANCE
```

The safeguard is the **terminal branch-length** quantile, the same value the terminal rule uses, not a quantile of ratios. A trivial branch in a short-branched neighbourhood can produce a huge ratio and is not a long-branch-attraction risk. If no valid quartet can be built, or the local reference still carries no scale, the taxon is retained and counted as not evaluable.

Both guards on the denominator exist because tree inference pins branches it cannot estimate at a minimum value.

Without them, ratios in the thousands appear - a first test run on real data spanned 7e-7 to 20805 against a median of 0.73. The Python prototype has no equivalent guard.

6. Everything that can block a removal

In every case the outcome is retention.

Rule	Condition	Result
all	unit observed in fewer than 3 trees (terminal, stem)	no trigger issued
all	3 gene trees or fewer in the dataset	terminal, stem and internal rules disabled
all	value exactly equal to a threshold	retained
stem	clade absent, or no monophyletic subset of 2 or more taxa	skipped
stem	taxa already covered by a larger removed clade	not reported again
internal	chain shorter than -p	no candidate
internal	outermost branch at or below the internal quantile	rescued
internal	side ratio at or below -r	blocked
internal	exact tie between the two side medians	blocked
internal	deeper side is not strictly smaller	blocked
quartet	fewer than 2 groups or fewer than 3 tips at every anchor	not evaluable
quartet	comparator closer to the anchor than 1e-6	not a valid comparator
quartet	local median at or below 1e-6	not evaluable

7. Parameter map

Flag / file	Position	Feeds
-t file	1	K1, terminal trigger
-t file	2	K2, internal trigger
-t file	3	K3, clade stem trigger
-t file	4	K4, quartet ratio threshold
-s file	1	q1, terminal quantile - used by the terminal AND quartet rules
-s file	2	q2, internal branch quantile
-s file	3	q3, clade stem quantile
-p	-	minimum branches in a long chain
-r	-	side asymmetry ratio, must exceed 1.0

8. Where the code lives

Function	Does
stats_for_terminals	per-taxon medians, M_terminal, triggers, pooled list
stats_for_stems	as above for clade stems; also collapses aliases per tree
stats_for_internal_branches	per-tree medians, M_internal, per-tree triggers
stats_for_quartets	the ratio distribution (used by the statistics script only)
resolve_clade_fragments	clade to fragment and stem length, in one tree
identify_long_terminals	rule 1

Function	Does
identify_clades_with_excessively_long_stems	rule 2, including alias and nesting
identify_long_branched_subtrees	rule 3, the four gates
clade_for_removal_by_asymmetry	gates 3 and 4
identify_local_long_branches	rule 4
build_quartet_mask_map, comparator_allowed	quartet masking
quartet_test_for_taxon	anchor search and the ratio

All of the above are in `Long_Branch_Remover/long_branch_rules.jl`. The libraries `TreesIO`, `TreeUtils` and `TreeStats` hold only general tree operations and no long-branch logic.

One sentence to remember. *Each rule asks whether a branch is long relative to its own kind (the trigger, K) and whether it is long in absolute terms across the dataset (the safeguard, q); both must hold, everything ambiguous is retained, and nothing is ever pruned - only listed.*